



A project report on

TWO WHEEL SELF BALANCING ROBOT FOR HOSPITALS

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KLE Technological University in partial fulfillment
of the requirements for the degree of

Bachelor of Engineering

in

Automation & Robotics

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Certificate

This is to certify that the project report entitled **Two Wheel Self Balancing Robot for Hospitals** is a bonafied work carried out by **Komal. P, Satvik. H, Shreya. V, Mansi. G** and **C. Rinku** in partial fulfillment for the award of the degree of Bachelor of Engineering in Automation & Robotics of KLE Technological University, Hubballi during the year 2023-2024. The project report has been approved in partial fulfillment for the award of said degree as per academic.

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ABSTRACT

Two-wheeled Self Balancing Robots in Hospitals

Recent years have seen tremendous developments in the application of robotic technology in healthcare settings, providing creative ways to improve patient care and optimize hospital operations. The design, development, and installation of a two-wheeled self-balancing robot designed primarily for hospital healthcare support are presented in this study. The main goal is to develop a flexible robotic platform that can help medical professionals, navigate hospital environments on its own, and increase overall efficiency. Key features of the two-wheeled self-balancing robot include obstacle avoidance and balancing. Additionally, the robot is equipped with a user-friendly interface, allowing healthcare professionals to interact with the system easily. The hardware architecture, control algorithms, sensory integration, and robot prototype of the two-wheeled self-balancing robot are among the technical components of this study that are investigated. The study also evaluates the robot's possible effects on patient care and workflow efficiency, as well as how well it performs in hospital. Keywords – Healthcare assistance, Hospitals, Autonomous navigation, Path planning, Obstacle Avoidance, Medical Supplies and medications.

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1. INTRODUCTION

1.1. BACKGROOUND

In today's world, devices like Segways and Hoverboards are commonly seen in movies, TV shows, YouTube videos, and similar platforms. The principle of a self-balancing robot, which constantly corrects its position to maintain balance on two wheels, is similar to these devices. The concept of mimicking the behavior of an inverted pendulum dates back to the '80s when a Japanese professor built the first model. Since then, numerous prototypes have been developed.

Self-balancing robots have become a captivating project for makers and students due to the widespread availability of economical electronic components. Unlike an ordinary pendulum that swings around a point with its center of gravity below it, a self-balancing robot has its center of gravity above the pivot point, making it inherently unstable. By carefully balancing the inverted pendulum, a very stable self-balancing robot can be created, which can stand still if the amplitude of the pendulum's oscillation is minimized.

Inverted pendulums, and by extension self-balancing robots, have a wide range of applications, including:

1. Segways
2. Rockets and missiles
3. Hoverboards
4. Toys
5. Goods Transportation
6. Surveillance

This report focuses on the design, construction, and control of a self-balancing robot specifically for hospital applications. Utilizing MATLAB for simulation, we employ complex mathematical modeling and control system concepts to manage the robot's stability and movement.

1.2. RELEVANCE OF PROJECT

The use of robotic technology in healthcare has become an influential force in recent years, bringing with it creative ways to improve patient care and maximize hospital operational effectiveness. The creation of a two-wheeled self-balancing robot designed especially for medical help is one such achievement. The purpose of this project is to present the robot's capabilities, guiding principles, and possible uses in medical environments. The two-wheeled self-balancing robot is a flexible platform that can move around hospitals' dynamic, sometimes busy environments on its own. Its ability to adapt to changing situations is improved by the integration of machine learning algorithms, which guarantees dependable and secure functioning.

The main goal of this study is to examine the difficulties that healthcare workers encounter when performing daily duties such as the delivery of prescription drugs, medical supplies, and diagnostic equipment. Healthcare personnel can refocus on more important parts of patient care by giving the self-balancing robot these tasks, which will ultimately increase workflow efficiency. This report provides the robot's technical details, such as its control algorithms, hardware architecture, sensory integration, and physical prototype. The robot functions in hospitals, assessing its capacity for independent navigation, obstacle avoidance, and real-time interaction with its surroundings.

1.3.PROBLEM STATEMENT: DESCRIPTION OF THE PROBLEM ADDRESSED

To design and curate a two-wheeled Self-balancing robot which is an automated robot that balances itself without any outside help or any support by their independently actuated wheels. Basically, it is about designing a robot for hospitals which can recognize the voice of the operator, locate a path, picks up the objects, and is supposed to complete the task in 5-7 minutes. So that is can assist healthcare professionals in a hospital or clinical setting by quickly responding to the instruction and retrieving objects and performing tasks such as medication delivery or waste collection.

Problem Statement: To design a Two-wheel self-balancing in the healthcare industry. The robot maintains balance using the moment of the wheels and body.

OBJECTIVES

- The robot will navigate autonomously through hospital corridors, delivering goods and medical supplies without the need for constant human supervision.
- The self-balancing nature of the robot will allow it to move smoothly and efficiently, even in tight spaces, slopes and reducing the risk of collisions or accidents.
- By remotely automating the transportation tasks, hospital staff can be freed up to focus on more critical patient care activities, improving overall labor efficiency.
- The robot will be designed to operate reliably in a hospital environment, with robust control systems ensuring stable and consistent performance.

1.4. NEED ANALYSIS

Patient Interaction: It determines how the robot will interact with patients. And it will be used for assistance, monitoring, or transportation.

Mobility and Accessibility: It assesses the environment where the robot will operate. Consider factors such as doorways, elevators, and the need to navigate around medical equipment.

Sensor Integration: Select appropriate sensors for navigation, obstacle detection, and environmental monitoring.

User Interface: Determines how healthcare professionals will interact with the robot. This may involve developing a user-friendly interface for controlling the robot's movements and accessing data.

2. LITERATURE SURVEY

2.1. LITERATURE SURVEY

Table 2-1: Literature survey

Sl.No	Authors	Title of the research paper	Journal name & year of publication
1.	Zhang, Wanying Wang, Shengda Shi, Peng Shi	A Two-Wheeled Self-Balancing Robot with the Fuzzy PD Control Method	Mathematical Problems in Engineering, 2012
2.	Bulfin, Susan Fuller, Willa Grobbel, Claudia	Anne Boykin Institute for the Advancement of Caring in Nursing Use of Robots to Complement Caring Relationships in Nursing Position Paper	International Journal for Human Caring, 2019-12-01
3.	Noh, Samyeul Park, Jiyoung Park, Junhee	Autonomous Mobile Robot Navigation in Indoor Environments: Mapping, Localization, and Planning	2020 International Conference on Information and Communication Technology Convergence (ICTC), 2020-10-21
4.	Sedlar, Damir Bosnjak, Andrea	Design and control based on the concept of an inverted pendulum	2023 8th International Conference on Smart and Sustainable Technologies (SpliTech), 2023-6-20
5.	Mathew, Alex Ananthu, R Binsy, P Vahid, Abdul Thomas, Ciby Sidharthan, S	Design and control of a two-wheel self-balancing robot	IOP Conference Series: Materials Science and Engineering, 2021-03-01

This position paper expresses current and future concerns about the use of robots in nursing. This statement reflects integration of humanoid nurse robots (HNRs) in nursing education, practice, and research. Nurses must be directly involved in decisions about the design, implementation, and evaluation of the use of HNRs in healthcare. As questions arise about the ethical use of robots in nursing situations, caring theories will inform the implementation of HNRs in nursing to ensure we protect the safety and the dignity of those entrusted to our care.[2]

Developing an autonomous indoor mobile robot navigation system [3] from scratch is very difficult and it takes a long time. It requires a series of complex tasks such as handling given sensor inputs, calculating all the information needed for autonomous navigation, and controlling actuators required for movement. In this paper [3] is introduced mainly based on open source provided by the robot operating system. The presented system is capable of autonomously navigating an unstructured indoor environment avoiding collision with static or dynamic objects. To this end, the system consists of three main modules: mapping, localization, and planning. The mapping module builds a global map for an unknown environment using a simultaneous localization and mapping algorithm based on laser scanner data. The localization module estimates the mobile robot's pose within the prebuilt map by way of an adaptive Monte Carlo localization approach. The planning module builds a local cost map for collision avoidance, generates collision-free trajectories to reach a goal pose based on the cost map, and produces control commands to follow the trajectories. The presented system has been tested not only in simulation environments built in the Gazebo simulator but also in real environments utilizing the Jackal mobile robot, to validate its performance for autonomous navigation including collision avoidance.

This research was aimed at designing, developing and analyzing a self-balancing robot with the implementation of a novel state feedback for the PID controller using on-board state estimation [4]. Unlike most of the previous works on self-balancing robots, which involved state estimation using IMU and feedback from the actuator side using encoders, potentiometers, or tachometers; this research predominantly focused on implementing an

on-board state estimation algorithm to generate the feedback, thus eliminating the requirement of sophisticated actuators. Such a system would not only be cost-efficient but would also be more robust in terms of state estimation since the actuator feedbacks have a recurrently bad impression of misleading the controller. Subsequently, the robot was designed and a prototype of it was tested for its performance. The performance analysis of the robot exhibited the expected behavior in terms of stability and robustness of the control design.

Designing the self-balancing two-wheeled mobile robots [5] and stabilizing it has gained a lot of momentum in research areas. Instead of complex control systems that are mostly seen in these research papers this paper introduces a new design of two-wheeled robot structure that fulfils the required level of stabilization without any complex control system. To achieve this, the structure is designed with a vertical straight-line motion. After deriving the mathematical model, the model is simulated in Matlab Simulink. The simulation is carried out for step, ramp, and uneven terrain input conditions.

2.2.MARKET SURVEY

Promising insights on the possible deployment of a two-wheeled self-balancing robot in healthcare settings were obtained from the extensive market survey. A growing trend in the demand for robotic solutions to solve the changing issues faced by healthcare professionals was identified by a review of the present market landscape. Numerous robotic technologies are currently in use, with the majority of them concentrating on the automation of jobs like distribution and logistics within hospital facilities. Analysis of competitors revealed a market with a combination of well-known companies and cutting-edge innovations. While other rivals provided all-purpose robots for diverse industries, a niche for healthcare support was found to present a chance for uniqueness. Important participants in the healthcare robotics sector showed different levels of market share, and there was a noticeable interest in hospital automation solutions.



Figure 2-1: Two-wheeled self-balancing Robots in the market

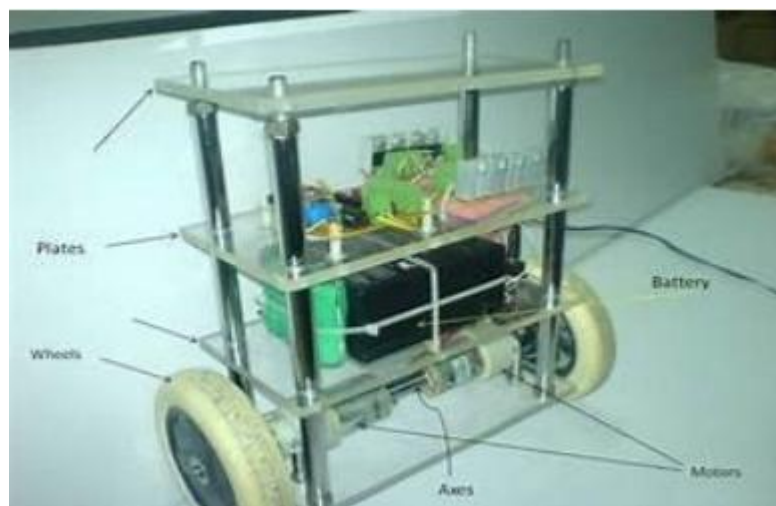


Figure 2-2: Robots available in the market

A favourable prediction was shown by a cost-benefit analysis, which linked possible cost savings to increased productivity, decreased labor needs, and simplified logistics. The expected benefits of improved patient care and operational efficiency seemed to outweigh the total cost of deploying the self-balancing robot. Evaluations of technology preparedness confirmed the feasibility of the self-balancing robot, which has strong hardware and smart software. The necessity for ongoing adjustment to changing hospital

surroundings and additional optimization of obstacle avoidance algorithms were among the recognized technological obstacles.

3. METHODOLOGY

The VDI 2206 methodology, also known as the "Association of German Engineers" (VDI) guideline for systematic development of mechatronic systems, is a structured approach used in the design and development of complex systems, including robotics. Here's an explanation of how this methodology might be applied to build a two-wheeled self-balancing robot system for hospital applications:

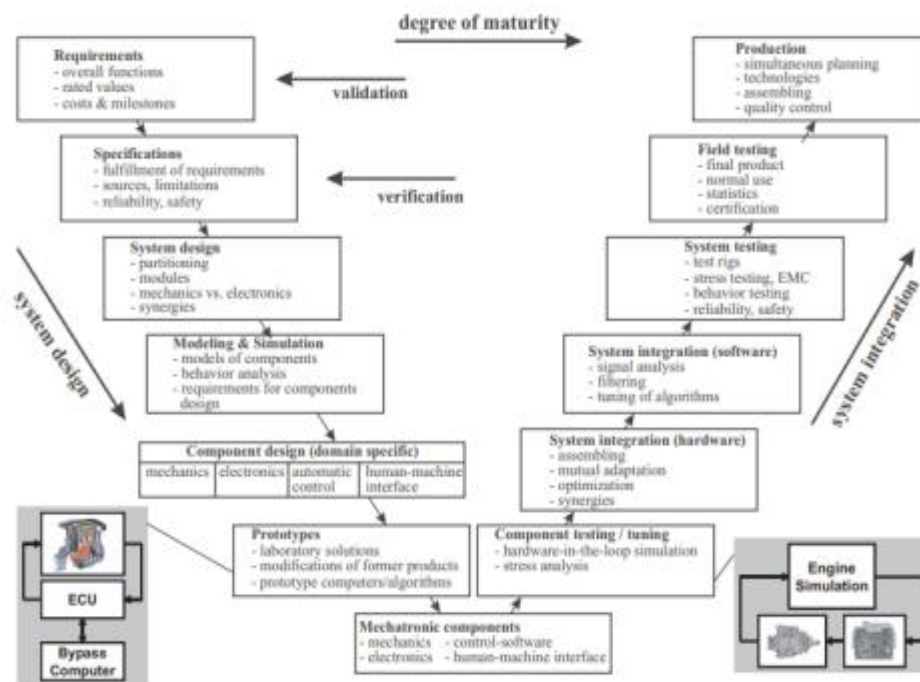


Figure 3-1: VDI 2206 Methodology

3.1. PERTINENT INFORMATION

With the continuous improvement of living standards and medical standards, the increased population has drawn great public attention. As the population is increasing, diseases, viruses, and infections are also increasing. Treating patients suffering from all these problems has become a big social issue. With the rapid development of

science and technology, our 2- wheel self-balancing medic bot has come into existence.

- Research on two-wheel self-balancing robots dates back several decades, but significant advancements have been made in recent years due to the availability of more sophisticated sensors, processors, and control algorithms. Some key areas of background research include:
- Control Systems: Researchers have developed various control algorithms to stabilize and control the movement of self-balancing robots. These include PID (Proportional-Integral-Derivative) controllers, LQR (Linear Quadratic Regulator) controllers, fuzzy logic control, and model predictive control.
- Sensor Integration: Early self-balancing robots relied on basic sensors like gyroscopes and accelerometers to measure orientation and acceleration. However, advancements in sensor technology have led to the integration of more precise and reliable sensors, such as IMUs (Inertial Measurement Units), encoders, and even vision systems for enhanced perception.
- Mechanical Design: The design of the chassis, wheels, and overall mechanical structure of self-balancing robots is crucial for stability, manoeuvrability, and efficiency. Researchers have explored different designs, materials, and configurations to optimize performance.
- Power and Actuation: Self-balancing robots require efficient power sources and actuators to maintain balance and move effectively. Research has focused on improving battery technology, motor efficiency, and power management systems to extend operating time and enhance performance.
- Applications: Beyond research and development, self-balancing robots have found applications in various fields, including personal transportation, surveillance and security, education and entertainment, industrial automation and medical sectors. Understanding the specific requirements and challenges in each application domain has driven further research and innovation.

3.2. REQUIREMENT ANALYSIS

Begin by thoroughly understanding the requirements of the hospital environment.

Identify the specific tasks the robot needs to perform, such as medicine delivery, logistics support, or patient interaction. Consider environmental constraints and safety requirements.

Functional requirements

- Balancing Algorithm.
- Motor control
- Safety features
- Sensor data collection and presentation
- Collision Detection
- Human-Machine Interface

System requirements

- The robot should operate in operation theatres and observation wards.
- The robot should self-balance.
- Easy and efficient user interface.
- It should provide clear and accurate sensor's feedback to operators.
- Communication
- Safety sensors

3.3.SPECIFICATIONS/ FUNCTIONS

- 2 wheeled medic-Bot Dimensions (L x W x H): 30 cm x 30 cm x 48 cm
- Minimum Ground clearance: 5cm
- Bot Weight: 2.5 to 3 kg (max)
- Pay-Load: 500g - 800g
- Operating Speed (Empty/Loaded): 150 mm/s, 115 mm/s
- Stop Position Accuracy: ± 20 mm

- Navigation: Remotely controlled / autonomous
- Collision: Beep sound
- Battery Life: Works for 3 hours once charge

3.4. QUESTIONNAIRE FOR STAKEHOLDERS

1. What specific tasks are beneficial in the healthcare environment?

- Medication and Supply Delivery and patient interaction

2. What safety features are crucial in a healthcare robot

- Collision Avoidance and emergency stop.

3. How can the robot efficiently integrate with existing healthcare workflows?

- The robot will be designed to work with adaptability with easy integration.

4. How will the robot manage to work during emergency times?

- The timing will be assigned to pick and give the object within the given time limit.

5. Can the robot move efficiently without any collision?

- Yes, it has navigation capabilities.

6. Are hygiene standards maintained?

- The robot's surfaces will be designed for easy cleaning and disinfection to maintain hygiene.

3.5. FLOW CHART

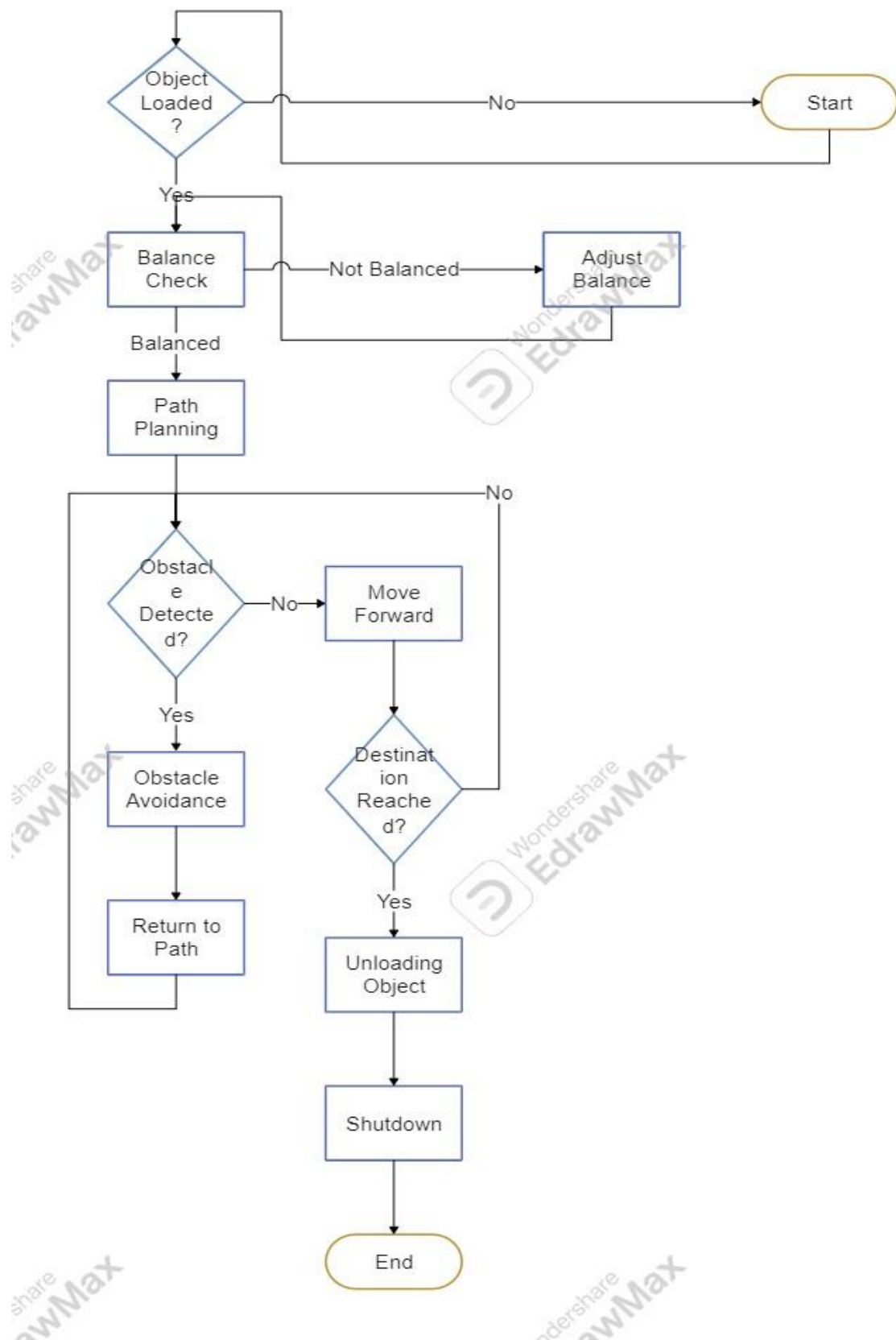


Figure 3-2: Overall functional flow chart of the robot

4. IMPLEMENTATION

4.1. FUNCTION TREE

Table 4-1: Function Tree

Balancing System	Sensor Data Collection	Data Processing	Control Algorithm
Movement System	Motor Control	Path Planning	Navigation
Load Handling System	Load Sensing	Load Stability	
User Interface	Input Methods	Feedback System	
Power Management	Battery Monitoring	Power Supply	

Balancing System: The balancing system of the two-wheel self-balancing robot consists of three main components: sensor data collection, data processing, and control algorithm. Sensor data collection involves gathering continuous data on the robot's angular velocity using gyroscopes and its acceleration and tilt using accelerometers. Data processing then combines the gyroscope and accelerometer data through sensor fusion for accurate tilt measurement. The robot's current tilt angle is computed from this fused data. The control algorithm employs Proportional-Integral-Derivative (PID) control to calculate the necessary adjustments to maintain balance. Based on these calculations, the robot adjusts motor speeds to stay upright.

Movement System: The movement system focuses on motor control, path planning, and navigation. Motor control involves adjusting motor speed to achieve the desired movement and changing motor direction for forward, backward, or turning motions. Path planning entails calculating the shortest and safest route to the destination, while obstacle detection and avoidance use sensors to identify obstacles and reroute as necessary. Navigation ensures the robot maintains a straight path within corridors and successfully navigates through doorways and into rooms.

Load Handling System: The load handling system includes load sensing and load stability. Load sensing measures the weight of the load being carried and ensures it is evenly distributed. Load stability involves adjusting the robot's posture to keep the load stable, particularly by managing the center of gravity. Shock absorbers are used to minimize load disturbance, ensuring the robot can carry medical supplies smoothly and safely.

User Interface: The user interface is designed for ease of use and feedback. Input methods allow users to control the robot manually using a remote through an application. The feedback system includes a status display, such as an LED or screen, to show the robot's operational status.

Power Management: Power management involves battery monitoring and power supply. Battery monitoring tracks the charge levels and power consumption of various components to ensure the robot operates efficiently. The power supply system includes a battery management system that optimizes battery use and longevity, as well as charging circuitry to manage charging operations safely and effectively. This ensures the robot remains operational and ready for use in hospital environments.

4.2. MORPHOLOGICAL CHART

Sub functions	Example1	Example2	Example3	Example4
Convert(motion to signal)				
Motor drivers				

Figure 4-1: Morphological chart



Figure 4-2: Morphological chart

4.3. CONCEPTUAL DESIGNS

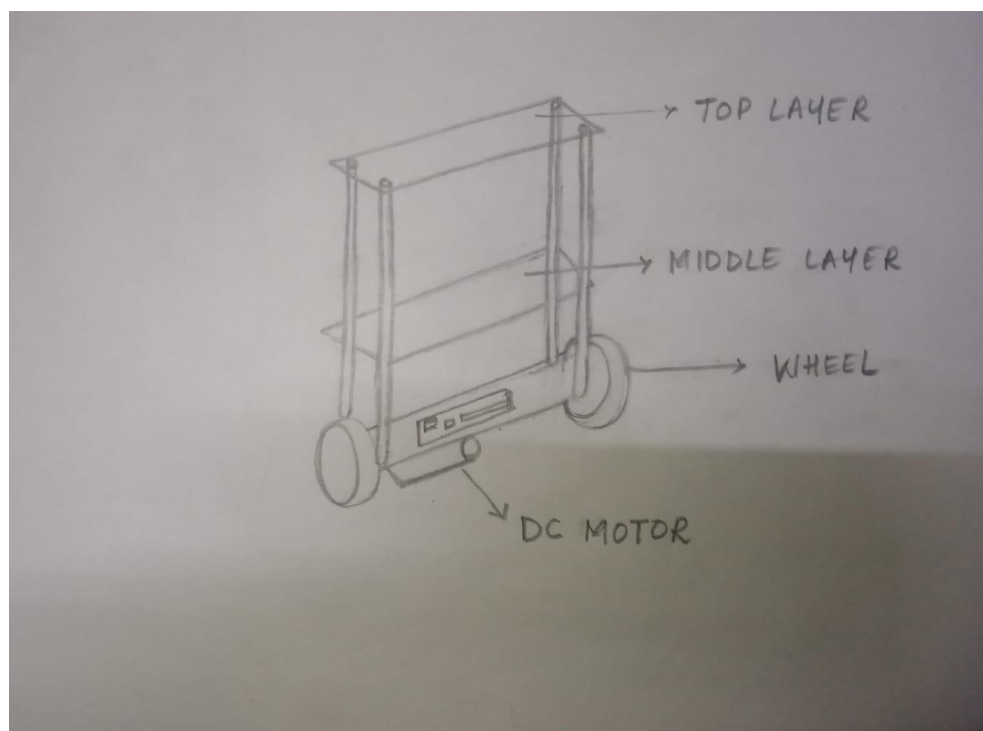


Figure 4-3: Conceptual Design 1

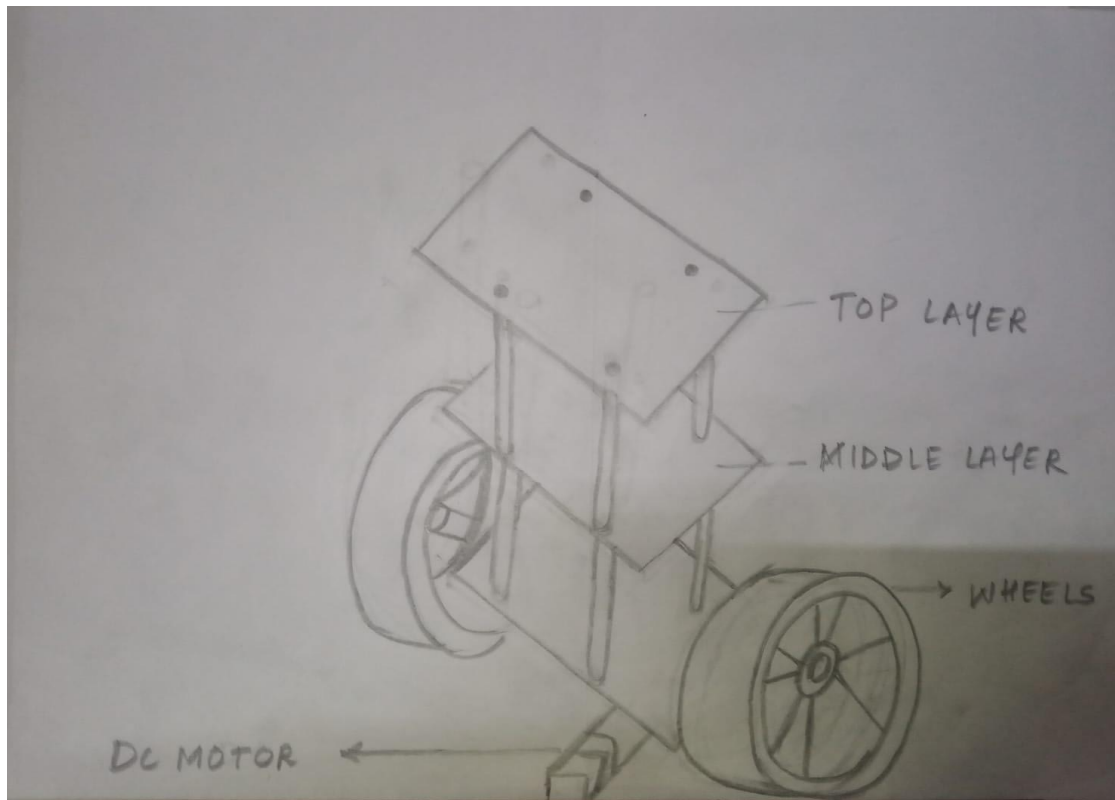


Figure 4-4: Conceptual Design 2

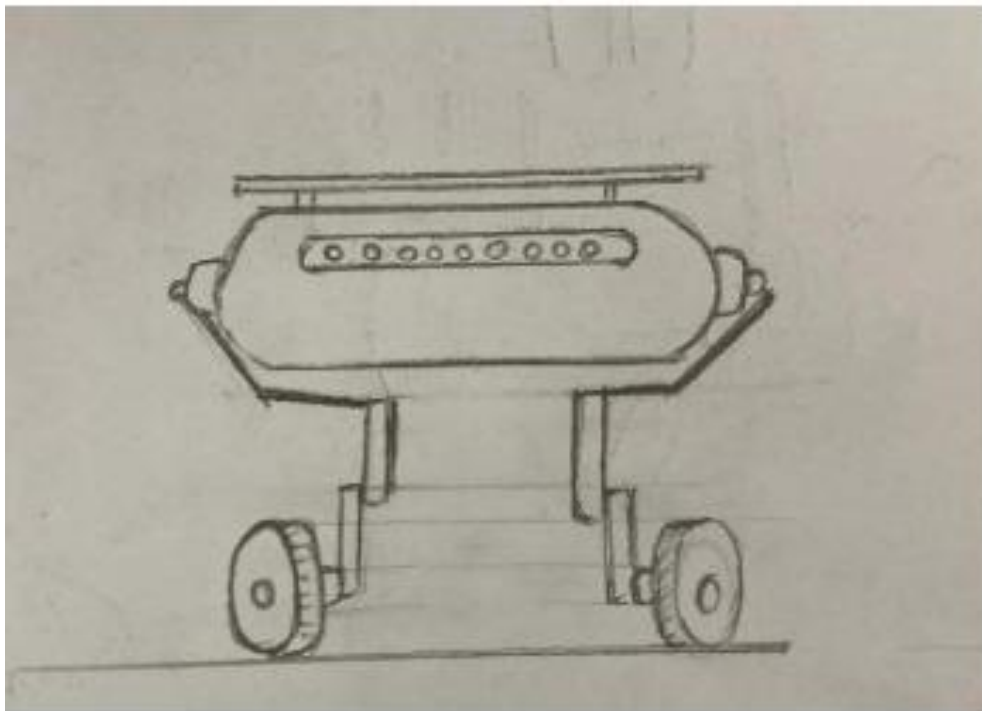


Figure 4-5: Conceptual Design 3

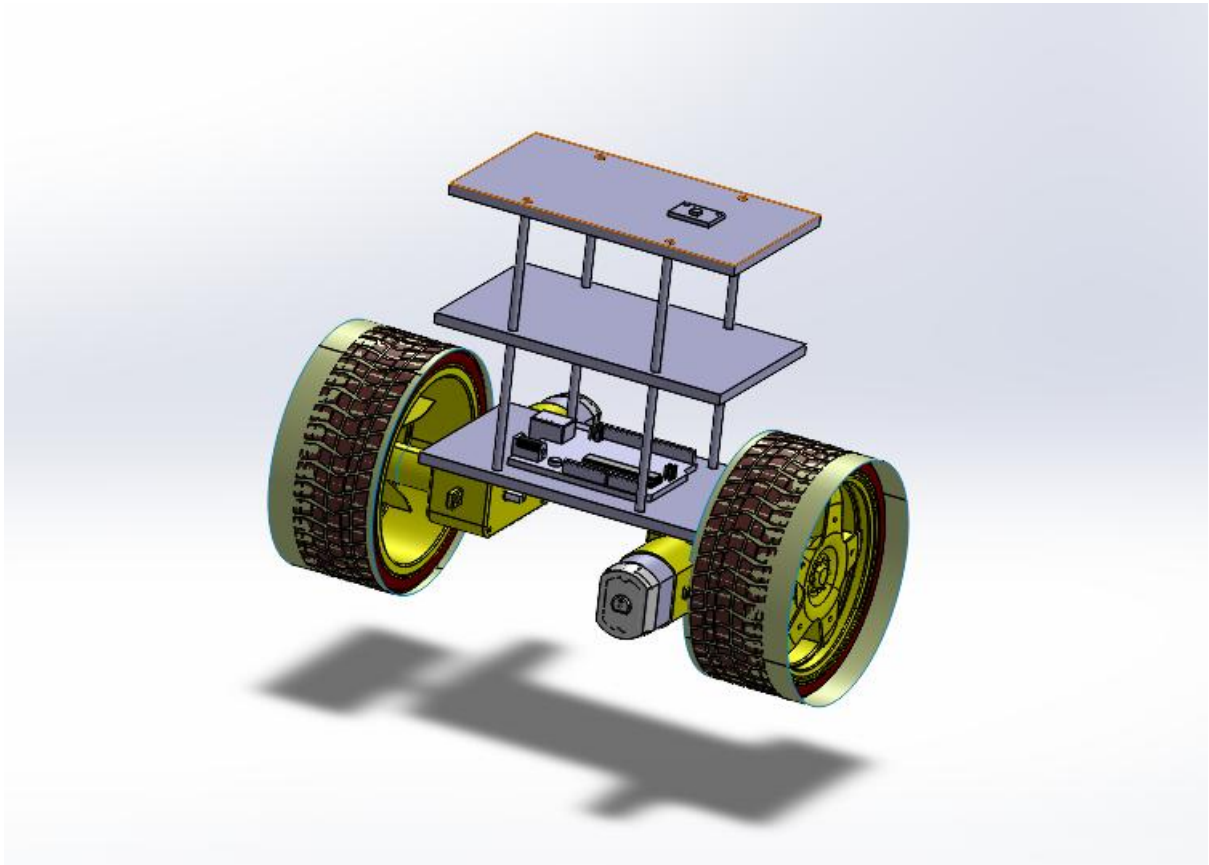


Figure 4-6: SolidWorks sketch indicating the body design of the two-wheeled Self Balancing Robot

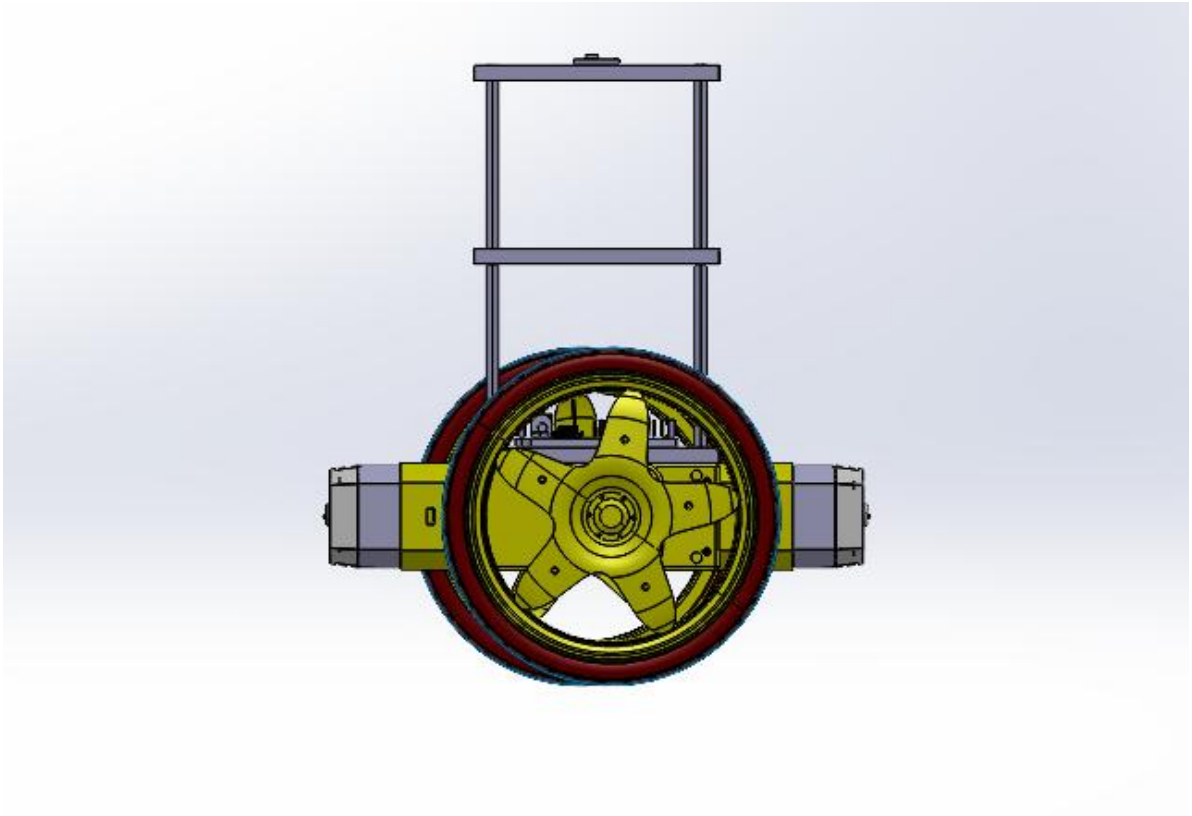


Figure 4-7: Side view of the Robot

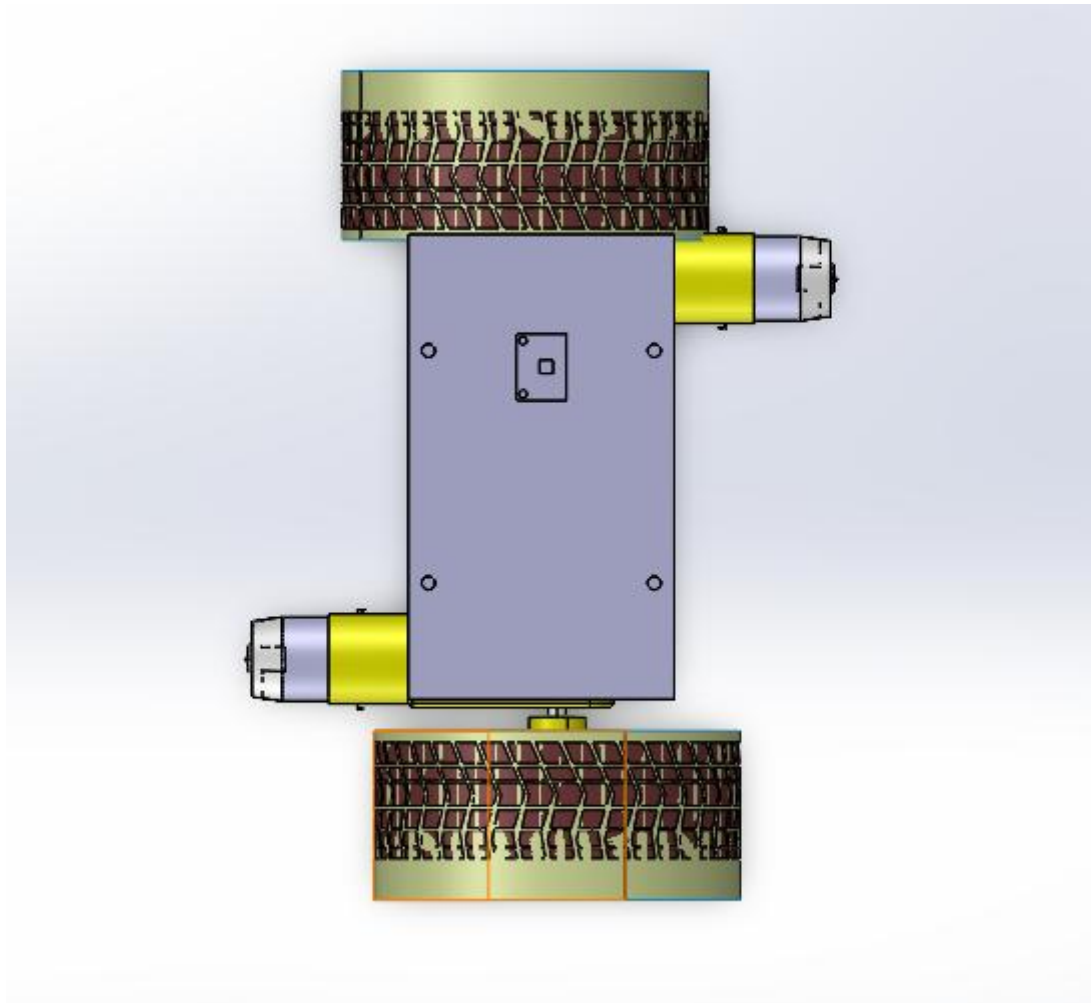


Figure 4-8: Top view of the robot

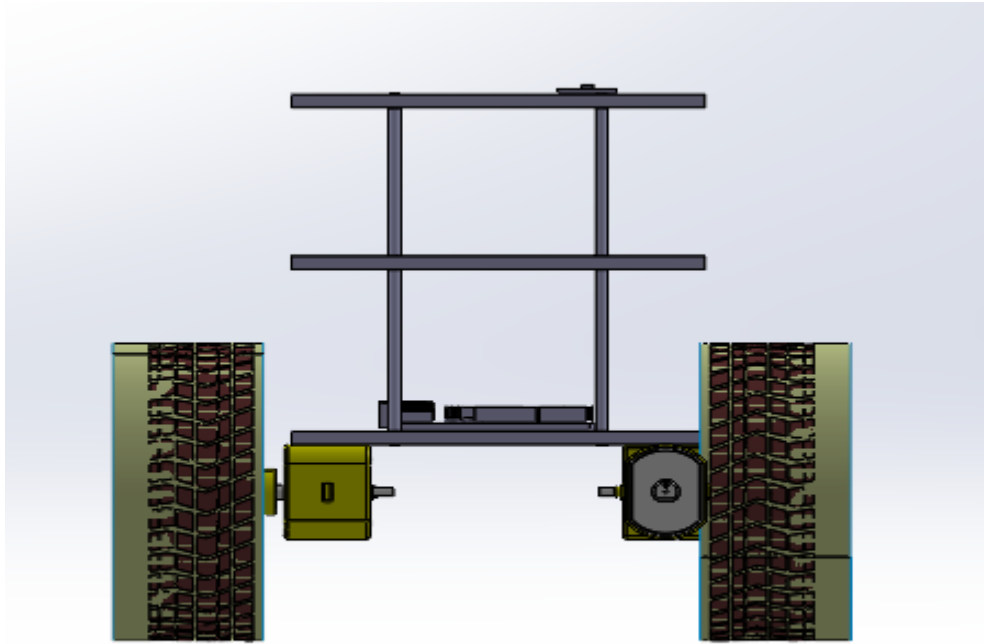


Figure 4-9: Front view of Robot

1. **Wheels:** The two large wheels are like the base of the robot, allowing it to move around. These wheels are crucial for balancing the whole structure. We have used plastic wheels of diameter 7cm x 2cm width.
2. **Body:** The platform you see stacked in layers to create shelves. This structure holds all the important components together. The body is of the dimensions 30cm x 30cm x 48cm (LxWxH) made up of Acrylic Sheets.
3. **Motors:** There are motors attached to the wheels, providing the necessary power to move and balance the robot. We have used 12V DC geared motors with encoders.
4. **Screws & Nuts:** These are used to fix the parts in place.
5. **Clamps & rods:** These are used for fixing the motors and provide support for the robot.

How It Works

- **Balance:** The robot needs to maintain its balance. The sensors detect any tilting, and the controller adjusts the motors to correct the tilt, keeping the robot upright.

- **Movement:** When the robot moves forward or backward, it slightly tilts in that direction. The motors then adjust to move it smoothly without falling over.

4.4. CONCEPTUAL DESIGN SCREENING

Table 4-1: NPS Score

Team No	08	Remarks
No of Promoters (9 to 10) =	10	As the NET promotor score is negative, it indicates the need for development.
No of Passives (7 to 8) =	10	
No of Detractors (0 to 6) =	15	
Net promotor score =	$29.40\% - 44.11\% = -15\%$	

4.5. FBS MODEL

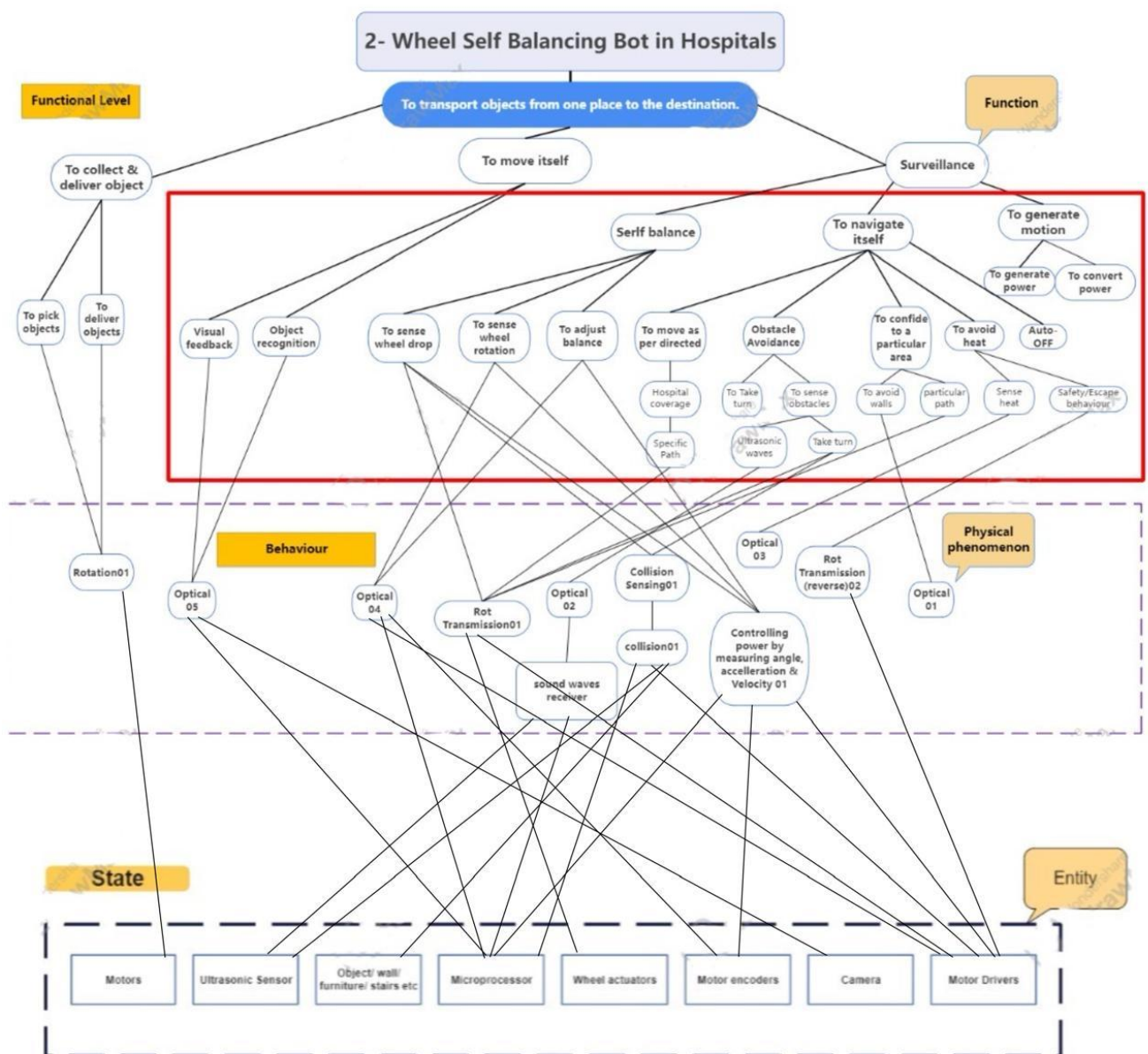


Figure 4-10: FBS model

The Function-Behavior-Structure (FBS) model depicted here illustrates the design and operation of a two-wheel self-balancing bot used in hospitals. This framework categorizes the system's aspects into functions, behaviors, and structures, offering a comprehensive understanding of how the bot works. At the highest level, the bot's primary function is to transport objects within a hospital. This involves sub-functions such as collecting and

delivering objects, moving autonomously, navigating obstacles, and generating the necessary motion through power management and conversion.

In terms of behavior, the bot must maintain its balance on two wheels. This involves using visual feedback from cameras or sensors, recognizing objects in their environment, sensing wheel drops, measuring rotational movement, and adjusting its position to stay balanced. The bot also needs to follow specific paths, avoid obstacles, stay within designated areas, and manage its power systems effectively, including an auto-off feature for safety.

The physical phenomena involved include various types of rotations and movements, collision sensing, and power control to maintain balance and facilitate movement. Optical phenomena, such as using light-based sensors for navigation and object recognition, are also crucial.

The state of the bot encompasses its physical components: motors for movement, ultrasonic sensors for obstacle detection, a microprocessor for data processing and control, wheel actuators for driving the wheels, motor encoders for measuring wheel rotation, and a camera for visual feedback and object recognition. Motor drivers control the power supplied to the motors.

The connections between these elements are depicted with arrows, indicating relationships and dependencies. Functional needs lead to specific behaviors, behaviors are achieved through physical phenomena, and the states describe the conditions and components involved in these processes.

4.6. SYSTEM DECOMPOSITION

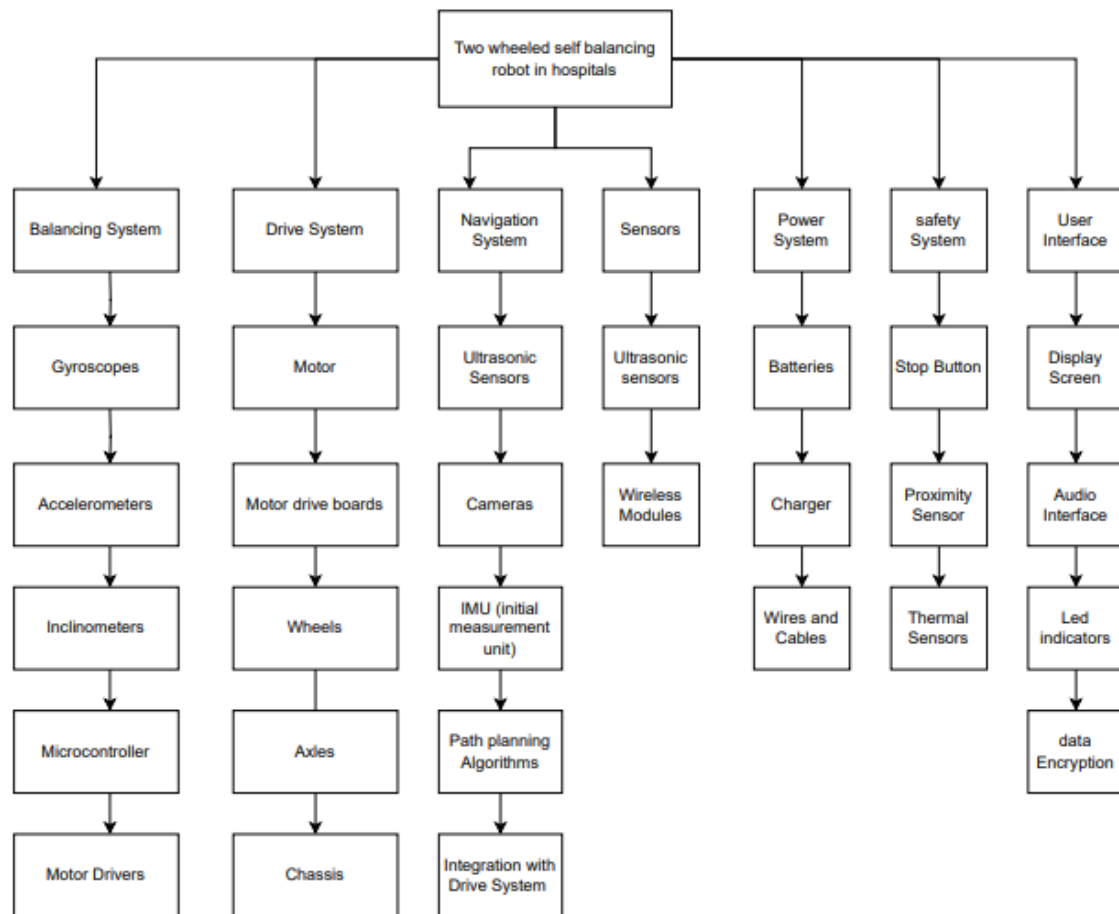


Figure 4-11: Robot system decomposition

4.7. INTERACTION OF SUBSYSTEMS

SI NO	SUBSYSTEM LIST
1	Balancing System
2	Drive System
3	Navigation System
4	Sensors
5	Power system
6	Safety system
7	User Interface

Figure 4-12: Subsystem List

The system decomposition for a two-wheeled self-balancing robot in hospitals is detailed in several interconnected subsystems, each fulfilling specific roles to ensure the robot's functionality and efficiency. This decomposition breaks down the robot into its core systems: the Balancing System, Drive System, Navigation System, Sensors, Power System, Safety System, and User Interface.

Balancing System

The Balancing System is crucial for maintaining the robot's stability on two wheels. Key components include gyroscopes, accelerometers, and inclinometers, which provide the necessary data on orientation and motion. These sensors feed information to the Raspberry Pi 4, which processes the data and sends commands to the motor drivers to adjust the motors and maintain balance.

Drive System

The Drive System consists of the components that enable the robot to move. This includes motors, motor drive boards, wheels, axles, and the chassis. The motor drivers receive signals

from the Raspberry Pi 4 to control the motors, facilitating movement and manoeuvrability. The chassis provides structural support, housing the drive system components securely.

Navigation System

The Navigation System ensures the robot can move autonomously within the hospital environment. This system incorporates ultrasonic sensors and cameras to detect obstacles and gather environmental data. An Inertial Measurement Unit (IMU) assists in tracking the robot's position and orientation. Path planning algorithms process this data to chart safe and efficient routes, integrating with the Drive System to guide the robot's movements.

Sensors

The robot is equipped with various sensors that support its functionality and safety. Ultrasonic sensors detect obstacles and measure distances, while cameras provide visual feedback for navigation and object recognition. Additional sensors include proximity sensors for nearby object detection.

Power System

The Power System supplies the necessary energy to the robot's components. It includes batteries, a charger, wires, and cables. The power system ensures that all subsystems receive a consistent power supply, enabling uninterrupted operation. Wireless modules may also be part of the power system, providing connectivity for remote monitoring and control.

Safety System

Safety is paramount in a hospital environment. The Safety System includes a stop button for immediate shutdown, proximity sensors to avoid collisions, and thermal sensors to prevent overheating. This system ensures the robot operates safely around patients and hospital staff.

User Interface

The User Interface facilitates interaction between the robot and human operators. It includes a display screen for visual output, LED indicators for status updates, a Cloud interface, and data encryption for secure communication. This interface allows users to monitor and control the robot, ensuring it performs its tasks effectively.

4.8. MAPPING

PHYSICAL EFFECT	COMPONENT	FUNCTION
BALANCING SYSTEM	Chassey and wheels	The body itself constitutes of chassey and wheels which will balance the overall bot.
DRIVE SYSTEM	Motor Drivers and IMU	Controls the motor speed and direction and the IMU consisting accelerometers and gyroscopes to measure the orientation and angular velocity.
NAVIGATION SYSTEM	Gyroscopes and Accelerometer	Measures the rate of rotation around the robot's axes in determining the robot's tilt angle by providing data on the gravitational force.
SAFETY SYSTEM	Battery and Power Management	Providing feedback on the position and speed. The feedback systems adjust motor performance and detect anomalies.
USER INTERFACE	Web Interface	Allows users to control the robot's movement remotely.

Figure 4-13: Mapping

4.9. PROJECT COMPARISON TABLE

Project Comparison Table-

Function	Sub - function	Robot 1	Robot 2	Robot 3
Balancing and motion	- Navigation - Random path - Obstacle detection - Speed monitor - Velocity controller	X	X	X
Localization	- GPS - Indoor localization system - Waypoint navigation		X	
Physical design	Enclosures and casings	X	X	X

Testing	To ensure stability and responsiveness	X	X	X
Energy Efficiency features	- Sleep Mode - Regenerative braking		X	
Audio feedback	Providing robot's status		X	
Data Logging	Recording sensor data			
Firmware management	Updates		X	X

Figure 4-14: Identification of similarities and differences

4.10. CALCULATIONS

As per stated requirements:

Bot weight = ~ 2kg

Payload = ~500 to 600 g

Wheel diameter = ~10 cm

Speed (M) = 25mm/s

Total mass = W+L = 2000+500 = 2500g

Assuming small friction = 0.1N

Torque(f) = 0.05 * 0.1 = 0.005Nm

Torque(acc) = $R * S * F = 0.05 * 25 * 0.1$ (radius*speed*force)

Therefore:

Torque = 0.125Nm

Hence total torque = $T = T_a + T_f$

$T = 0.05 + 0.125$ $T = 0.175$ Nm

In terms of KG-cm

$T = 1.75$ kg-cm

4.11. PRODUCT SYSTEM ARCHITECTURE

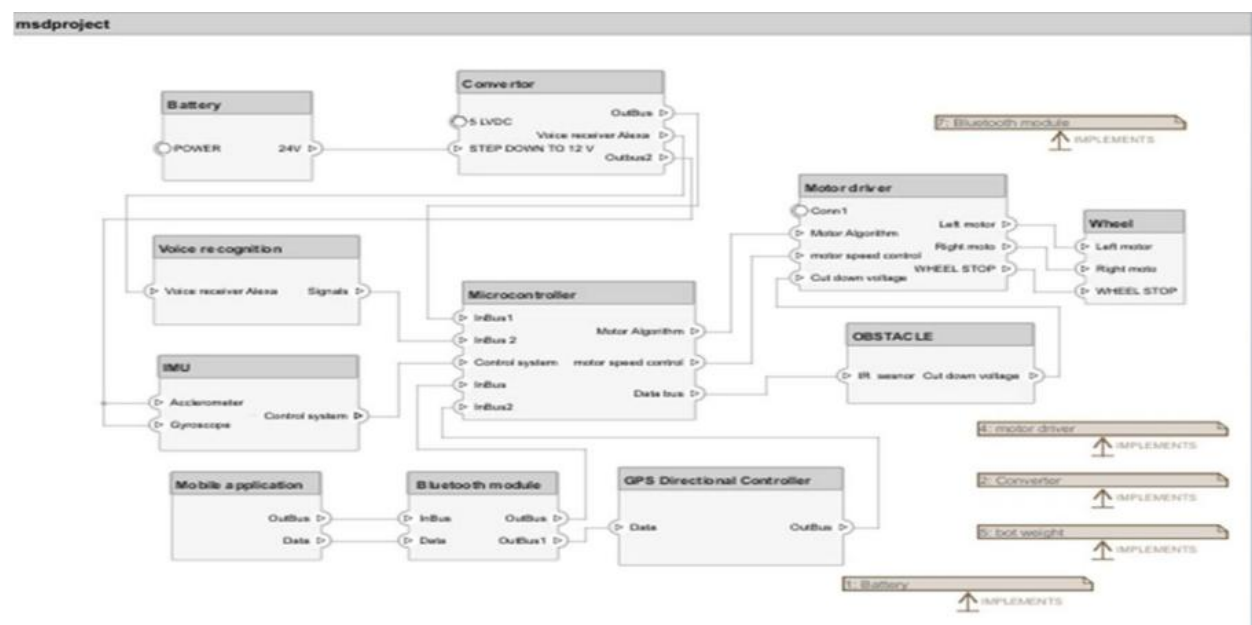


Figure 4-15: System composer architecture

4.12. SELECTION OF EMBEDDED SYSTEM

Raspberry Pi: - is an embedded Linux system. It is running on an ARM and can give us the experience of using an embedded design. Raspberry Pi OS (formerly known as Raspbian) is a Unix-like operating system based on the Debian GNU/Linux distribution for the Raspberry Pi family of compact single-board computers.

4.13. TOOLCHAIN

1. MATLAB - used for PID tuning and model simulation using it .
2. SOLIDWORKS - used for designing the bot (3D).
3. Raspberry Pi imager - used for writing the OS into the memory card.
4. Putty - used to connect to the pi.
5. Real VNC viewer - used to get the remote desktop of the pi, for reasons such as coding and changing certain requirements.
6. Thonny - text editor for typing and running commands.

4.14. SELECTION OF COMPONENTS

Sensor Selection: -

1. Ultrasonic sensor HC-SR04: - This sensor uses ultrasonic waves to detect objects in front of it. So, we are incorporating this sensor in our project as it serves the purpose of obstacle detection.
2. MPU-6050 (Gyroscope & Accelerometer): - This sensor aids in the measurement of velocity, orientation, acceleration, displacement, and other motion-related features.
3. So, we use this to get the desired orientation of the bot.

Motors and Drivers Selection: -

SPG30E: - 200K DC Geared Motor with Encoder

Rated Voltage: 12 V

No-load Current: 70 mA

Rated Current: 410 mA

Rated Torque: 1.8 kg-cm

Shaft Diameter: 6 mm

L298N: - is the motor driver we used for driving the motors

4.15. CIRCUIT DIAGRAM

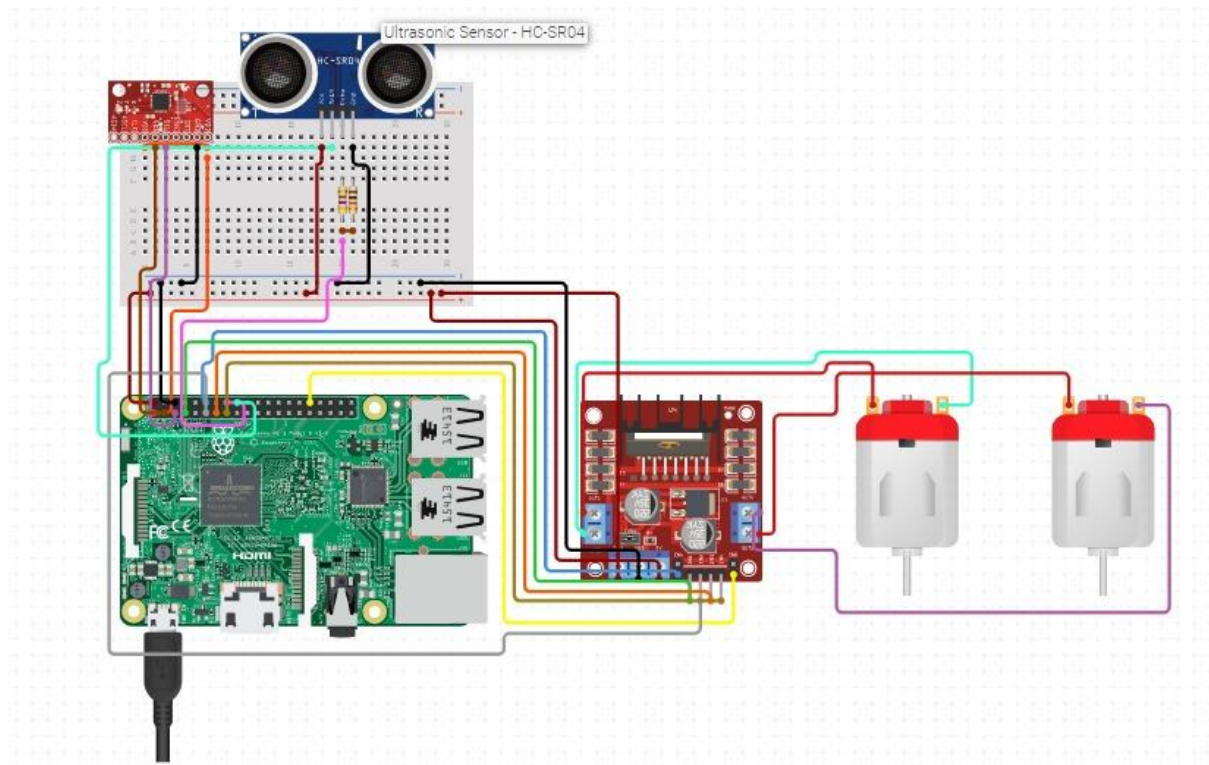


Figure 4-16: Circuit Diagram of Robot

This circuit diagram illustrates the connections for a 2-wheel self-balancing robot, incorporating a Raspberry Pi, an ultrasonic sensor (HC-SR04), a motor driver (L298N), a Camera module, and two DC motors. The Raspberry Pi acts as the primary controller for the robot, interfacing with both the ultrasonic sensor, camera and the motor driver.

The ultrasonic sensor (HC-SR04) is connected to the Raspberry Pi with four wires: the VCC wire (red) connects to the 5V power on the Raspberry Pi, the GND wire (black) connects to the GND, the TRIG wire (green) connects to GPIO 23, and the ECHO wire (yellow) connects to GPIO 24.

The motor driver (L298N) has several connections. For the power supply, the VCC wire (red) connects to a 12V power source, while the GND wire (black) connects to both the GND on the Raspberry Pi and the power source. The motor driver controls two motors. For Motor A, the OUT1 wire (pink) and OUT2 wire (purple) connect to the two terminals of the left motor. For Motor B, the OUT3 wire (light blue) and OUT4 wire (green) connect to the two terminals of the right motor. The control pins on the motor driver are connected to various GPIO pins on the Raspberry Pi: IN1 (orange) to GPIO 5, IN2 (brown) to GPIO 6, IN3 (yellow) to GPIO 13, IN4 (blue) to GPIO 19, ENA (white) to GPIO 12, and ENB (gray) to GPIO 18.

The two DC motors with encoder are connected to the motor driver (L298N), which drives the robot's wheels.

Raspberry Pi Camera Module is connected to the CSI (Camera Serial Interface) port on the Raspberry Pi, which is typically located near the HDMI port.

4.16. HARDWARE ARCHITECTURE

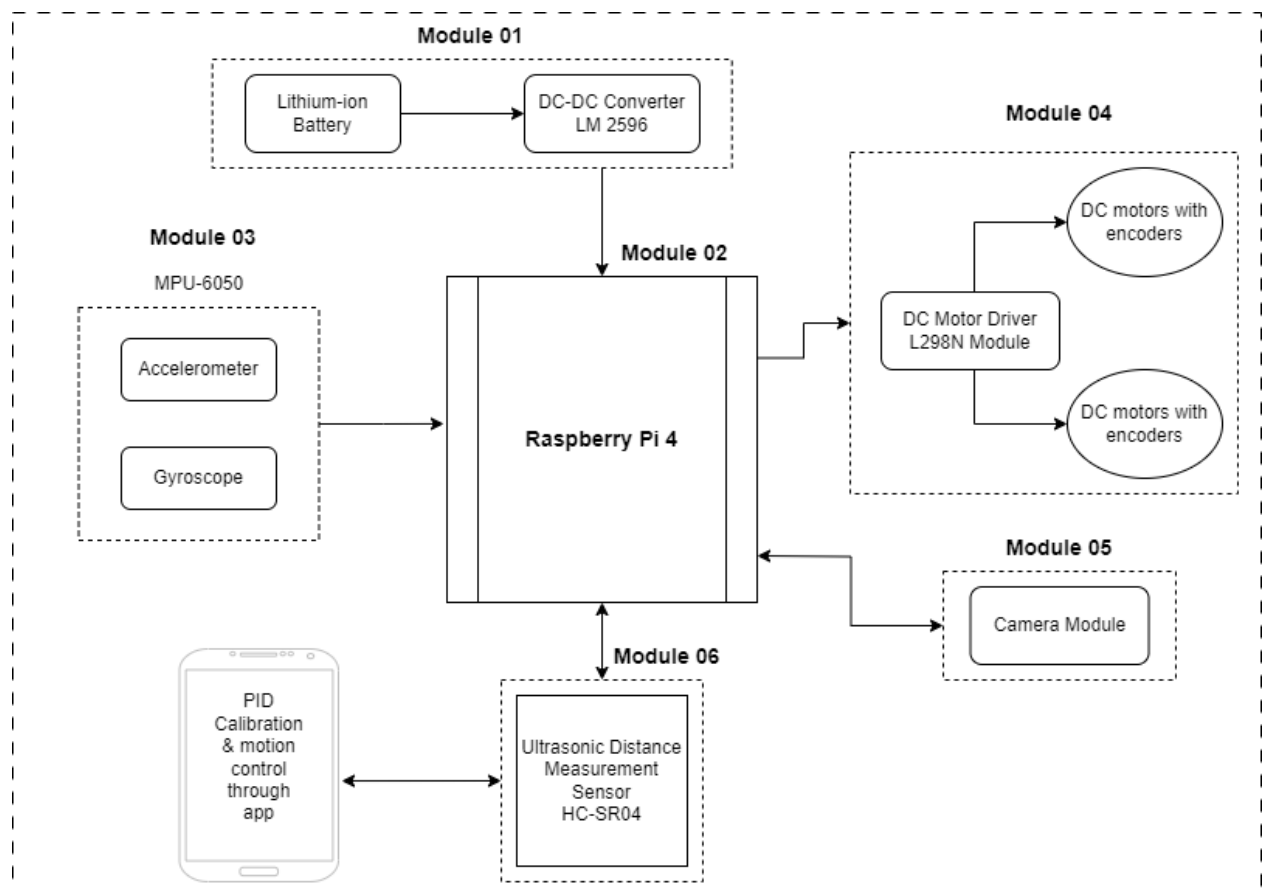


Figure 4-17: Block diagram

This block diagram provides an overview of a 2-wheel self-balancing robot, detailing the interconnected modules and their functions. The central component of the system is the Raspberry Pi 4, which serves as the main controller and is responsible for processing data and controlling the robot's movements.

Module 01 comprises a Lithium-ion battery connected to a DC-DC converter (LM2596). This module supplies power to the entire system by converting the battery's voltage to a suitable level for the Raspberry Pi and other components.

Module 02 is the Raspberry Pi 4 itself. It acts as the brain of the robot, receiving inputs from various sensors and controlling the motor driver to maintain balance and move the robot.

Module 03 includes the MPU-6050 sensor, which contains both an accelerometer and a gyroscope. These sensors provide crucial data on the robot's orientation and motion. The accelerometer measures the linear acceleration, while the gyroscope measures the angular velocity. This information is sent to the Raspberry Pi for processing to ensure the robot remains balanced.

Module 04 consists of the motor driver (L298N) and DC motors with encoders. The motor driver receives control signals from the Raspberry Pi and adjusts the speed and direction of the DC motors accordingly. The encoders on the motors provide feedback on the motor's position and speed, allowing for precise control.

Module 05 is the Raspberry Pi Camera Module connected to the Raspberry Pi. This module enables the robot to capture images or video, which can be used for various purposes such as navigation, object detection, or surveillance.

Module 06 involves the ultrasonic distance measurement sensor (HC-SR04). This sensor measures the distance to obstacles in front of the robot by emitting ultrasonic waves and calculating the time it takes for the waves to return. This information is sent to the Raspberry Pi to help avoid collisions.

Lastly, the diagram shows a smartphone with an application used for PID calibration and motion control. The app communicates with the Raspberry Pi, allowing the user to fine-tune

the robot's balancing parameters (Proportional, Integral, and Derivative) and control its movements remotely.

4.17. BOM

Table 4-2: BOM

Electronic Components			
Sl.No	Component Name	Specification	Quantity
1	Raspberry Pi	4 B	1
2	Accelerometer & Gyroscope	MPU-6050 (2.375V–3.46V)	1
3	Motor driver	L298N (12V)	1
4	DC motor with encoder	JGA25-370 280/ 220 rpm (12V)	2
5	Ultrasonic Sensor	HC-SR04 (5V)	1
6	Regulator module	LM2596 DC-DC (3.3-V, 5-V, 12-V)	1
7	External Camera module	Raspberry Pi Camera (8 Mp)	1
8	Resistor	470 ω , 1k ω	1
9	Jumper wires	-	As required
Physical Material			
1	Plastic wheels	diameter 7cm x 2cm width (for 4mm dia motor shaft)	2
2	Acrylic sheets	22mm x 12mm (4mm thickness)	3
3	screws & nuts	-	As required
4	metal rod	5mm dia	As required

4.18. BUDGETING AND PROCUREMENT

Table 4-3: Budgeting and Procurement List

Sl.no	Item name	Budgeting	Procurement
1	Acrylic sheet		Provided by college
2	Raspberry pi	8000/	Provided by college
3	Motor driver	290/	Procured from an electronic store
4	Motors (SPG30E- 200K DC)	1800/	Provided by college
5	Ultrasonic Sensors	180/	Procured from the electronic store
6	Pi Camera module	380/	Procured from the electronic store
7	Plastic Wheels	350(for 2)/-	Procured from the electronic store
8	Screws, nuts & clamps	150/-	Procured from the electronic store

4.19. ACTION PLAN FOR FABRICATION OF PROTOTYPE

Creating a prototype of a 2-wheel self-balancing robot involves a series of detailed steps, from initial planning and design to the final assembly and testing. The process begins with careful planning and design, where the robot's dimensions and overall structure are defined using CAD software. This phase also includes identifying all necessary components, such as the Raspberry Pi, sensors, motors, and batteries, and determining their placement within the acrylic structure.

Next, the material preparation phase involves acquiring the required acrylic sheets, as well as other necessary materials like screws, nuts, spacers, and mounting brackets. Ensuring all electronic components, such as the Raspberry Pi, MPU-6050, HC-SR04, L298N motor driver, and DC motors, are available and ready for assembly is also crucial at this stage.

The fabrication of the acrylic parts follows, utilizing both laser cutting and traditional tools like drills and saws. Laser cutting is employed for intricate designs due to its precision and clean cuts. This involves transferring CAD design files to the laser cutter software, setting up the machine, and executing the cutting process. For larger, less intricate parts, saws and drills are used to cut and create holes, ensuring all edges and holes are smoothed for a clean finish.

Once the acrylic pieces are cut, they are assembled into the robot's chassis using screws, nuts, and spacers. It is essential to ensure all parts fit together securely, providing a stable structure. This assembly phase includes test fitting all electronic components to verify their alignment with the mounting holes.

Mounting the electronic components onto the acrylic chassis is the next step. The Raspberry Pi, MPU-6050, HC-SR04, and camera module are attached to their designated positions, and the L298N motor driver is secured along with the DC motors with encoders. Proper routing and securing of wiring are critical to avoid interference and ensure reliable connections.

Following the physical assembly, the wiring and connections phase ensures that all electronic components are correctly connected and powered. This involves connecting the power supply from the DC-DC converter to the Raspberry Pi and motor driver and verifying all sensor and motor driver connections to the appropriate GPIO pins on the Raspberry Pi.

With the hardware setup complete, the software installation and testing phase begins. This includes installing the Raspberry Pi OS and necessary libraries for the camera and sensors, uploading control algorithms to the Raspberry Pi, and calibrating the MPU-6050 sensor.

Testing the motors and verifying their response to control signals are also conducted during this phase, along with fine-tuning the PID parameters using a smartphone app.

Final adjustments and thorough testing follow, where the robot's self-balancing capability is tested and necessary adjustments are made to the code or hardware. Obstacle avoidance tests using the ultrasonic sensor, camera functionality checks, and a series of test runs are performed to verify the robot's stability and performance.

The process concludes with documentation and review, creating detailed records of the design, fabrication, and testing processes.

5. RESULTS

5.1. WORKING PROTOTYPE

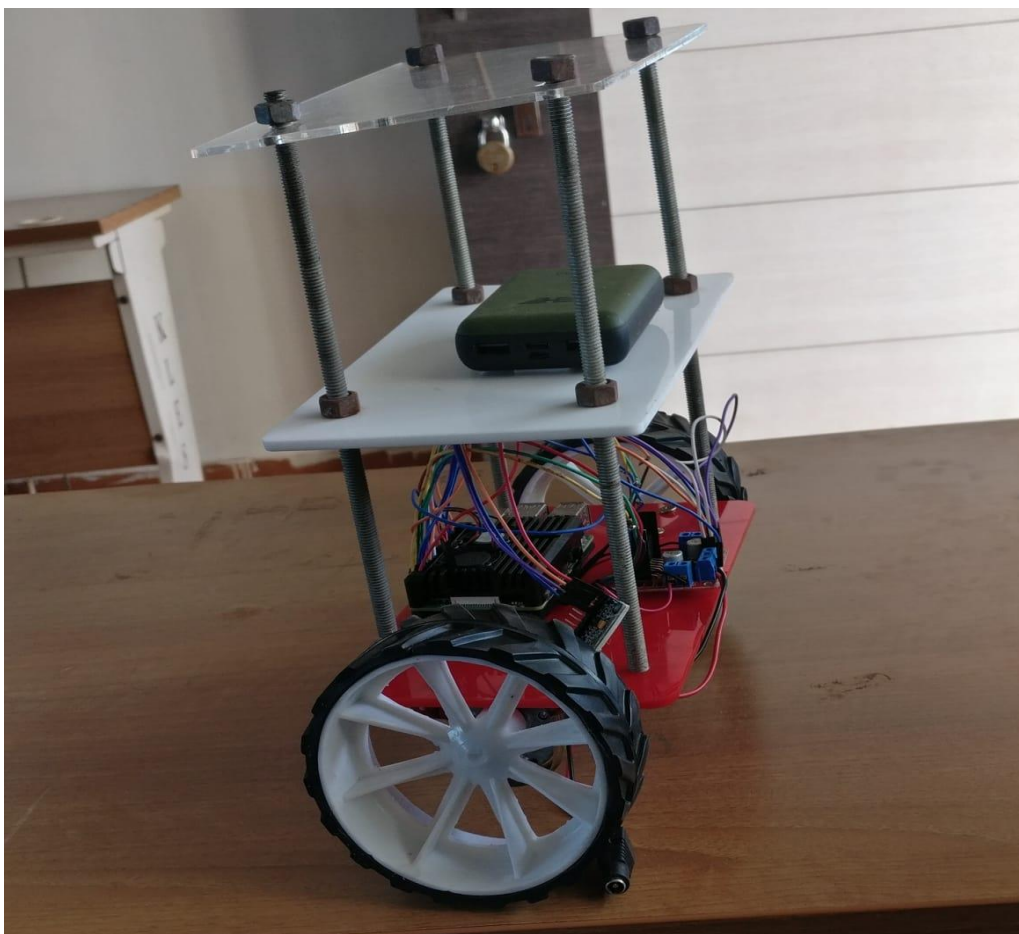


Figure 5-1: 2-wheel self-balancing robot physical prototype

5.2. TESTING OF PROTOTYPE

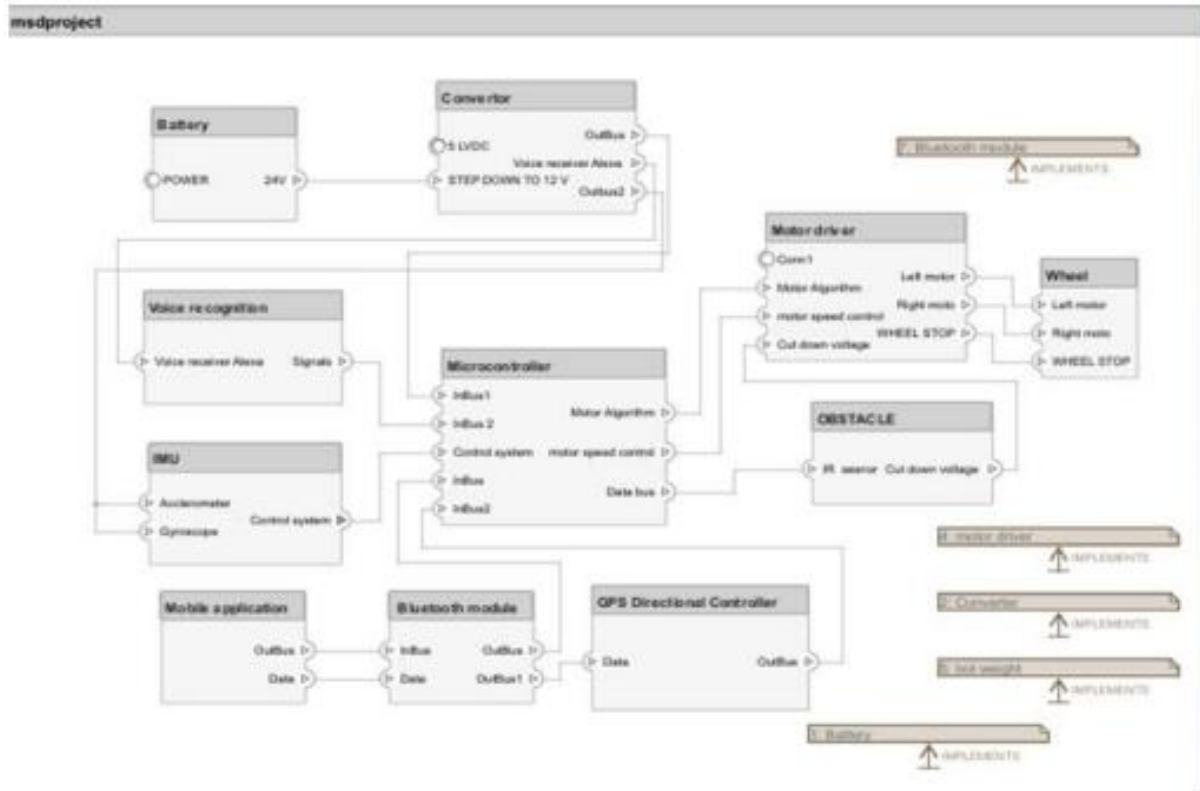


Figure 5-2: MATLAB toolbox

5.3. USER FEEDBACK ON PROTOTYPE

Table 5-3: Feedback

Team No	08
Mode of collecting feed back:	Asked Professors, project guide and Seniors.
Significant Feed back points for your prototype	1. Stability and balance 2. obstacle detection and avoidance 3. Load capacity 4. Battery life and charging 5. Maintenance
Your learnings for further improvement of your prototype	1. The self balancing robot should be more user interfaced And should be more acquire more accuracy for transportation of objects. 2. Automated path planning and integration of ROS.

6. CONCLUSION AND FUTURE WORK

The two-wheeled self-balancing robot for healthcare assistance holds great promise in addressing critical challenges within the healthcare sector. The innovative design principles, rooted in a user-centric approach, aim to create a robot capable of autonomously navigating hospital environments, minimizing infection risks, and enhancing operational efficiency.

6.1. CONCLUSIONS

The incorporation of advanced sensors, such as gyroscopes and accelerometers, signals a commitment to enabling autonomous navigation with obstacle avoidance, crucial for the robot's safe and efficient movement within dynamic healthcare spaces. The emphasis on infection control, achieved through minimizing human contact with medical supplies, aligns with current healthcare priorities and emphasizes the potential of the robot to contribute to a safer healthcare environment. The user interface design, a pivotal aspect of the concept, focuses on creating a seamless interaction between healthcare professionals and the robot. This user-friendly approach aims to facilitate easy integration into existing workflows, ultimately enhancing the overall user experience. The conceptual design also recognizes challenges such as algorithm fine-tuning and regulatory compliance. These challenges,

viewed as opportunities for improvement, set the stage for ongoing refinement in future design iterations, ensuring adaptability to evolving healthcare needs. The robot is capable of navigating in all sorts of pathways including slopes and inclined ways.

6.2. FUTURE WORK AND SCOPE

The future work and scope of the 2-wheel self-balancing robot in hospitals project are vast and promising, with potential applications in various aspects of healthcare delivery and hospital operations. As technology continues to advance, the integration of such robots into hospital environments can significantly enhance efficiency, safety, and patient care.

One of the primary future works involves enhancing the robot's autonomy and intelligence. This can be achieved by incorporating advanced sensors, machine learning algorithms, and computer vision capabilities. These improvements would enable the robot to navigate complex hospital environments more effectively, avoiding obstacles, recognizing and interacting with patients and staff, and making real-time decisions. The development of robust navigation and localization systems is crucial for the robot to operate autonomously, and perform tasks such as delivering medications, transporting medical supplies, and assisting in telemedicine consultations.

Another area of future work is improving the robot's communication and integration with existing hospital systems. By equipping the robot with advanced communication protocols and interoperability standards, it can seamlessly integrate with electronic health records (EHRs), hospital information systems (HIS), and other digital infrastructure. This would allow the robot to access and update patient information, schedule tasks, and coordinate with other hospital devices and systems, ensuring smooth and efficient operations.

The scope of deploying 2-wheel self-balancing robots in hospitals is extensive. These robots can be used to enhance patient care by providing companionship and assistance to patients, especially those with limited mobility. They can also help reduce the workload of hospital staff by taking over repetitive and time-consuming tasks such as delivering meals, cleaning, and monitoring patients. In emergencies, these robots can act as first responders, providing immediate support and assistance until human medical personnel arrive.

Additionally, the integration of telemedicine capabilities within these robots can revolutionize remote healthcare. Equipped with cameras and communication tools, the robots can facilitate virtual consultations between patients and doctors, enabling remote diagnosis and treatment. This is particularly beneficial in scenarios where patients are in isolation wards or when there is a shortage of medical staff.

The future work also includes exploring the potential of these robots in specialized hospital units such as operating rooms, intensive care units (ICUs), and rehabilitation centers. In operating rooms, robots can assist surgeons by providing tools, maintaining a sterile environment, and even performing precise surgical tasks. In ICUs, they can monitor patients' vital signs and alert medical staff in case of emergencies. In rehabilitation centers, robots can assist patients in performing exercises and track their progress, providing valuable data to therapists.

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